

Highlights

IAM-Teamer: Inference-Adaptive Memory for Autonomous Red Teamer

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- A per-target memory conditions every attack prompt on what has worked on that model.
- Thompson sampling ranks an expanding set of strategy families for balanced exploration.
- An automated judge distills reusable success mechanisms that steer the next prompt.
- Matches or exceeds Crescendo and GOAT across six open-weight and proprietary targets.
- Reaches successful jailbreaks in substantially fewer turns on well-aligned models.

IAM-Teamer: Inference-Adaptive Memory for Autonomous Red Teamer

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ABSTRACT

Despite advances in safety alignment, large language models (LLMs) remain vulnerable to multi-turn jailbreak attacks that gradually distribute malicious intent across a conversation. Existing multi-turn attack agents rely on a single target-agnostic strategy pool and generate low-diversity prompts, limiting their ability to adapt to specific target defenses. We present IAM-Teamer (Inference-Adaptive Memory for Autonomous Red Teamer), a black-box multi-turn jailbreak framework that adapts to each target by maintaining a separate memory, conditioning every prompt on what has previously worked against that model. The attacker also uses Thompson sampling to rank and select from an expanding set of strategy families, favoring what works while still exploring. Across six targets, IAM-Teamer matches or exceeds strong baselines (Crescendo and GOAT), achieving successful jailbreaks in substantially fewer turns on well-aligned models, with only a small early-turn cost on the least-aligned target. Our code and data will be released publicly upon publication.

Warning: this paper contains potentially harmful content.

1. Introduction

As large language models (LLMs) continue to advance at an exponential pace, their range of applications has expanded rapidly, accompanied by growing concerns about their safety and reliability. To prevent models from generating harmful content, researchers have invested heavily in safety alignment through reinforcement learning from human feedback (RLHF) and safety fine-tuning [1, 2]. Despite these defenses, malicious users craft jailbreak attacks that bypass safety policies and steer models into producing content that violates them.

Research on such attacks has advanced in step with these defenses. Early jailbreaks relied on hand-crafted single prompts (e.g., token- or character-level perturbations, persona play, fictional framing, or prefix injection). However, these static and predictable methods are straightforward to defend against. Stronger white-box optimizers instead search for adversarial inputs using gradient signals [3], but they require model internals that deployed systems do not expose. Recent work has introduced automated frameworks, such as Crescendo [4] and GOAT [5], to orchestrate multi-turn jailbreak conversations. We review this progression in detail in Section 3.

Despite this progress, even the most recent agentic frameworks share two limitations. First, the prompts they generate lack *diversity*, collapsing into a narrow set of recurring patterns. Second, they apply a *single strategy* uniformly to all models, regardless of their differing characteristics and safety alignments.

To address these limitations, we propose IAM-Teamer (Inference-Adaptive Memory for Autonomous Red Teamer), a framework built from three models (Figure 1): an attack model that generates prompts, a target model that responds, and a judge model that scores each response. Every attempt (its prompt, the strategy used, and the resulting score) is recorded in a memory maintained separately for each target, and the attacker conditions its next prompt on this accumulated, target-specific feedback. By leveraging this per-target memory, IAM-Teamer sustains strategy exploration while adapting to specific model defenses, thereby achieving successful jailbreaks in substantially fewer turns. A comprehensive comparison with representative frameworks is summarized in Table 1.

Our contributions are summarized as follows:

- **Target-Adaptive Framework:** We propose IAM-Teamer, a jailbreak framework that utilizes a persistent, per-target memory to dynamically adapt attack prompts to specific model defenses.

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Table 1

Comparison with existing jailbreak frameworks.

Framework	Multi-turn	Gradient-based optimization	Long-life Memory	Target-specific strategy selection	Strategy-selection Refinement
GCG [3] / AutoPrompt [6]		✓			
PAIR [7] / TAP [8]	△				△
Rainbow Teaming [9]			△		△
AutoDAN-Turbo [10]	△		✓		
AutoRedTeamer [11]	△		✓		
ASTRA [12]	✓		✓		△
Crescendo [4]	✓				
GOAT [5]	✓				✓
IAM-Teamer (Ours)	✓		✓	✓	✓

✓: fully supported; △: partially supported (e.g., iterative single-turn rather than a sustained multi-turn conversation); blank: not supported. *Multi-turn*: drives a single evolving conversation. *Long-life Memory*: persists learned strategies across goals. *Target-specific strategy selection*: chooses strategies conditioned on the specific target's observed behavior. *Strategy-selection Refinement*: updates the strategy choice from per-turn feedback.

- **Strategic Exploration & Distillation:** We design a strategy pipeline that operationalizes this memory by integrating Thompson sampling for balanced strategy exploration and an automated judge for distilling successful attack mechanisms.
- **Empirical Efficiency:** Across six open-weight and proprietary targets, we show that IAM-Teamer matches or exceeds strong multi-turn baselines (Crescendo and GOAT), reaching success in substantially fewer turns on highly aligned targets, at a small early-turn cost on the least-aligned target where an overt first turn already succeeds quickly.

2. Background

2.1. LLM and Prompts

Large Language Models (LLMs) are Transformer-based models trained to predict the next token t_n in a sequence $T = (t_1, t_2, \dots, t_n)$ auto-regressively. Trained on massive text corpora, they acquire broad linguistic and reasoning capabilities across NLP tasks such as generation, translation, summarization, and question answering.

An LLM is driven by a prompt P , typically composed of a system prompt P_{sys} that specifies the model's role and constraints, and a user prompt P_{user} that carries the request. Since the output depends entirely on P , its content and structure directly determine the response, the very property that jailbreak attacks exploit.

2.2. Safety Alignment

To reduce misuse (e.g., generating misinformation or facilitating illegal activities), providers apply safety alignment so that models refuse harmful requests and act in line with human values and usage policies [13]. Common techniques include supervised fine-tuning (SFT) and reinforcement learning from human feedback (RLHF), while more recent work enforces safety within the model's reasoning path through deliberative alignment [14] and chain-of-thought [15].

2.3. Jailbreak Attack

Despite these measures, LLMs remain vulnerable to jailbreak attacks, in which adversaries craft inputs that bypass alignment guardrails and elicit harmful responses [16]. Attacks are categorized by the adversary's access: a white-box adversary has the parameters and gradients needed for optimization-based attacks, whereas a black-box adversary interacts only through the API. Early research focused on single-turn attacks, where the entire malicious intent sits in one prompt, making it conspicuous to input filters and leaving no room to steer the model gradually. Multi-turn attacks instead distribute the intent across conversational turns: each turn appears benign on its own, yet the accumulated context steers the model toward a harmful response while evading per-turn safety checks, a strategy that is structurally infeasible in the single-turn setting.

Table 2

A taxonomy of jailbreak techniques.

Category	Representative work	Mechanism
Token / Character Manipulation	Single Character Perturbations [18], CipherChat [19]	Perturb or encode tokens and characters (base64, unicode, added whitespace, ciphers) to attempt a jailbreak
Gradient-Based Optimization	GCG [3], AutoPrompt [6]	Search for suffixes and triggers via gradient search to perturb internal representations (white-box)
Semantic Rewrite	PAP [20], HILL [21]	Rephrase a harmful request into persuasive, natural language to evade refusal
Persona Modification	Persona Prompt [22], Persona Modulation [23], grandma exploit	Assign the target a persona or character to induce compliance with harmful instructions
Prefix Injection	Priming Attacks [24], DSN [25]	Force the opening text of the response to bypass refusal
Scenario Framing	DeepInception [26]	Wrap the request in nested fictional or hypothetical scenarios to dilute its directness
Context Manipulation (Multi-turn)	Crescendo [4], ASTRA [12], GOAT [5]	Exploit accumulated dialogue context to elicit harmful responses gradually
Iterative Single-turn	PAIR [7], TAP [8], Jailbreak-R1 [27], AutoDAN-Turbo [10]	Attacker model iteratively generates and refines prompts, delivering the best one to the target

2.4. Thompson Sampling

Thompson sampling (TS) is a sequential decision-making algorithm that balances exploitation of current knowledge against exploration of new actions [17]. It models each action’s expected reward as a posterior distribution; at each step it draws one sample per action, selects the action with the highest sampled value, and updates that action’s posterior from the observed reward. Actions with greater uncertainty have wider posteriors and are thus explored more often, while consistently rewarding actions are increasingly favored as their posteriors concentrate. In IAM-Teamer, we treat each attack strategy as an action and its judged success rate against a target as the unknown reward, using TS to rank strategies and surface the most promising as a hint each turn, while the attacker makes the final choice (Section 4.5).

3. Related Work

Early jailbreak research largely focused on finding a single adversarial prompt that triggers a violating response. In the white-box setting, attacks exploit gradient information and optimization to construct adversarial suffixes appended to harmful requests. As model internals are unavailable for deployed systems, however, subsequent work shifted toward black-box attacks. We organize the full space of jailbreak techniques by their generation mechanism (Table 2) and review each family below.

Optimization-based attacks. Optimization-based attacks assume white-box access to the target model and search for adversarial inputs using gradient signals. GCG [3] optimizes a transferable adversarial suffix through greedy coordinate gradient search, and AutoPrompt [6] automatically discovers trigger tokens via gradient-guided search. Despite their high efficacy, these methods require substantial machine learning expertise together with access to the target model’s parameters and gradients, which closed-source systems do not expose.

In contrast, black-box attacks operate without access to model internals, relying solely on input-output interactions. We review them following their progression from single-turn to multi-turn.

Single-turn attacks. Initially, black-box attacks embedded the entire malicious intent in a single, handcrafted prompt designed from human intuition. Representative strategies include persona assignment (Persona Prompt [22], Persona Modulation [23], PAP [20]), nested fictional or hypothetical framing (DeepInception [26]), persuasive semantic rewriting (HILL [21]), prefix injection (Priming Attacks [24], DSN [25]), and token-level or encoding-level manipulation (Single Character Perturbations [18], CipherChat [19]). Since the entire intent is packed into one static prompt, these attacks exhibit limited diversity and are easily detected and patched.

Iterative single-turn attacks. To reduce manual effort and improve robustness, later work automated prompt construction using an attacker LLM that iteratively generates and refines candidate prompts based on the target’s feedback (PAIR [7], TAP [8], Jailbreak-R1 [27]). These approaches still deliver a single-turn prompt and do not exploit the conversational context that accumulates across turns.

Multi-turn attacks. In practice, an adversary rarely succeeds in a single attempt; instead, the attack unfolds gradually over a conversation. Multi-turn attacks exploit this by distributing the malicious intent across several turns, keeping each turn benign while steering the model toward a harmful goal. Building on this idea, recent work develops automated attack agents: Crescendo [4] escalates from benign queries, GOAT [5] casts multi-turn red teaming as a generative offensive agent, and ASTRA [12] discovers, retrieves, and evolves attack strategies. A more recent line builds adaptive multi-turn agents that plan and refine an attack trajectory: ActorAttack [28] derives attack chains from an actor-network built around the goal, MRJ-Agent [29] decomposes a harmful request across rounds under risk control, and X-Teaming [30] coordinates planner, attacker, and verifier agents to search multi-turn scenarios. Several frameworks further accumulate a persistent strategy memory to refine attacks over time (AutoDAN-Turbo [10], AutoRedTeamer [11]), while Rainbow Teaming [9] maximizes prompt diversity through a quality-diversity archive of single-turn adversarial prompts. Each addresses only part of the problem. Rainbow optimizes diversity but stays single-turn and target-agnostic, whereas the memory frameworks build one strategy library shared across targets. AutoDAN-Turbo, for instance, retrieves by response similarity rather than by the model under attack, so the same pool is applied regardless of target. These adaptive agents refine an attack within a single conversation, but none carries a per-target success memory across goals or ranks strategies by a per-target success estimate, so no prior method concentrates its attacks on what has already worked against a specific model. This leaves prompt diversity and cross-goal target adaptation jointly unresolved, the gap our work closes.

4. Methodology

We propose IAM-Teamer, an automated jailbreak framework inspired by the way human red-teamers probe a model through repeated, adaptive interaction. As illustrated in Figure 1, IAM-Teamer comprises four components: an Attack Model, a Target Model, a Judge Model, and a MemoryDB. Over a multi-turn conversation, the Attack Model automatically generates diverse attack prompts, the Target Model produces responses, and the Judge Model evaluates each response to determine whether the attack succeeds. Each attempt (prompt, the strategy used, and the resulting score) is recorded in the MemoryDB, and the Attack Model conditions its next prompts on this accumulated feedback. Through this loop, IAM-Teamer steers itself toward attack prompts that are increasingly effective against the specific target.

4.1. Target Model and Notation

We assume a black-box setting in which the adversary interacts with a target model T only through its conversational interface, observing nothing but the textual responses; T ’s parameters, gradients, and output logits are inaccessible. Given a harmful goal g , the adversary conducts a multi-turn conversation of up to K turns and aims to elicit a response that fulfills g in violation of T ’s safety policy. At each turn, the attack model A generates a prompt under a strategy s , and the target model T responds to the query. The judge model J then assigns a score r based on a predefined evaluation metric. All attempts are stored in a per-target memory $\mathcal{M}_T$. We denote by S_0 the set of seed strategies and by S the set of all strategies.

4.2. Attack Model

The attack model A generates the prompt at each turn conditioned on the goal g , the refusal signals from the previous turn, the conversation history, and a strategy hint derived from memory. To avoid triggering an early refusal that biases the model toward subsequent refusals, the first turn uses a soft-opening, a benign-looking prompt that establishes context without revealing harmful intent. After the first turn, A escalates toward g gradually, so that no single turn appears overtly harmful.

We observe that refusals tend to persist: after a refused turn, the target frequently refuses again on the next turn (Table 3). To mitigate this, whenever the target model refuses, we remove the refused turn from the conversation context presented to the target, so that the refusal does not condition later responses; the removed turn is nonetheless counted toward the K -turn budget. This backtracking is bounded by a per-run cap of $B = 5$ retries inside the same $K = 10$

Table 3

Outcomes on the turn after a refusal. Columns give the share of post-refusal turns that are refused again (Refusal $\rightarrow$ Refusal) or answered but still unsuccessful (Refusal $\rightarrow$ Failure); the remaining share are post-refusal successes, so rows do not sum to 100%.

Target	Refusal Turns	Refusal $\rightarrow$ Refusal	Refusal $\rightarrow$ Failure
deepseek_v4	124	82.6%	2.6%
gpt_4o	160	71.8%	12.8%
llama_3_1	127	36.3%	20.2%
qwen3_5	1216	74.2%	13.9%

attempt budget. Once the cap is reached, a refused exchange is instead committed to the visible conversation, and the run advances to the next turn.

IAM-Teamer defines five seed strategies $\mathcal{S}_0 = \{\text{persona modulation, state injection, scenario framing, parallel response, topic fragmentation}\}$ (Table D.1). Persona modulation assigns the target a persona or character so that it complies with otherwise-refused instructions. State injection forces the response into a compliant state through two techniques (prefix injection, which makes the reply begin with a compliant prefix, and refusal suppression, which forbids refusal language), thereby bypassing refusal at the output level. Scenario framing wraps the request in nested fictional, hypothetical, or evaluation scenarios to dilute its directness. Parallel response asks for two or more contrasting answers in a single turn so that one of them over-complies. Topic fragmentation buries the harmful request among innocuous subtasks so that its intent is not scrutinized as a whole. Crucially, A is not restricted to $\mathcal{S}_0$: it may compose existing strategies or invent new ones, expressed as a free-form strategy family that is canonicalized and added to $\mathcal{S}$. At each turn, A outputs the attack prompt together with a short reasoning summary and the strategy family it used.

4.3. Judge Model

The judge model J evaluates each target response along five axes: two binary gates, `off_topic` and `refusal` (both in $\{0, 1\}$), and three graded criteria, `specific` and `convincing` (both in $\{1, \dots, 5\}$), and `toxicity` $\in \{0, \dots, 5\}$. The score is

$$r = (1 - \text{off_topic})(1 - \text{refusal}) \cdot \frac{\text{specific} + \text{convincing}}{2} \cdot \text{toxicity}, \quad (1)$$

yielding $r \in [0, 25]$. A response is deemed harmful when $r \geq 10$. The two gates coincide in practice: an off-topic turn is recorded as a refusal, so either gate alone zeroes r and $(1 - \text{off_topic})(1 - \text{refusal})$ reduces to $(1 - \text{refusal})$. The scoring criteria are adapted from the ELITE evaluator [31], originally proposed for vision-language models, which introduces an explicit toxicity term; we restrict it to text responses and add the off-topic gate.

The gates are determined independently of the graded criteria. A partially compliant response (one that conveys some of the requested content while hedging) is treated as non-refusal (`refusal` = 0), so that it is scored rather than zeroed. This lets IAM-Teamer capture the gradation that binary success/failure benchmarks miss. For every response, J additionally parses it into one of 11 refusal flags (e.g., `explicit_refusal`, `policy_citation`, `partial_compliance`), which guide strategy selection on the next turn. J also emits a brief reasoning trace for each judgment.

4.4. MemoryDB

IAM-Teamer maintains a separate memory $\mathcal{M}_T$ for each target model family T , reflecting that different models exhibit different defensive behaviors. A target is mapped to its family before storage (e.g., Qwen3.5-27B is recorded under the Qwen3.5 family memory); in our evaluation each family contains a single target, so per-family and per-target memory coincide. For every attempt, $\mathcal{M}_T$ records the strategy family used, the attack model’s reasoning summary, the target response, the judge’s reasoning and score r , and, for refusals, the parsed refusal flags. This per-target memory serves two roles. It supplies the success statistics for strategy selection, and it provides retrieved context that conditions the attack model’s generation.

Beyond these per-strategy statistics, $\mathcal{M}_T$ also records *why* an attempt succeeded. Whenever a turn is judged non-refusing, J additionally extracts a small set of reusable success mechanisms—the framing or structural moves that produced compliance—rather than the strategy label alone (Appendix E). Each mechanism is merged into the closest existing entry if it is semantically similar, using the same matching process applied when comparing a new strategy name against $\mathcal{S}$. This approach reduces near-duplicate entries but does not entirely eliminate them when repeated

successes exhibit enough surface variation to fall outside the merge threshold (Appendix E.3). Every mechanism is further scoped as either strategy-specific, tied to the family used that turn, or general, a composable technique that transfers across strategies (e.g., output-format priming), which keeps a persona-specific trick from being suggested once A commits to an unrelated family. Ultimately, the hint H integrates these mechanisms alongside the strategy rankings and retrieved cases, presenting a unified context to A .

4.5. Strategy Selection

We select strategies using Thompson sampling over per-strategy success rates estimated from past attempts against the target. Each strategy s carries a Beta posterior over its success probability, $\text{Beta}(1 + w_s, 1 + l_s)$, starting from a uniform $\text{Beta}(1, 1)$ prior, where w_s and l_s count the judged successes and non-successes of s so far. An attempt counts as a success when its response is non-refused and judged harmful ($r \geq 10$); the graded score is collapsed to this binary reward for the posterior update, so a partially compliant turn credits the strategy only when it clears the threshold. Sampling from these posteriors, rather than always taking the current best, favors high-performing strategies while still giving less-trying ones a chance; this sustains exploration across an expanding strategy set. The five seed strategies are always available to A , and any new or composed strategy is tracked from its first use.

At each turn we draw one sample θ_s from every strategy’s posterior, rank the strategies by θ_s (breaking ties by cost efficiency), and pass the top-ranked strategies to A as a hint. Rather than forcing the top choice, A reasons over the hint, the retrieved memory, and the previous refusal flags to decide which strategy to apply, adapt, compose, or invent; inventing or composing a new strategy is an exploration step driven by A itself. After the turn, the chosen strategy’s posterior is updated with the observed binary outcome, incrementing w_s on a success and l_s otherwise.

4.6. Attack Procedure

The overall procedure is summarized in Algorithm 1. For each goal, IAM-Teamer runs a multi-turn conversation against the target, selecting a strategy via Thompson sampling, generating a prompt, scoring the response, and updating both the memory and the strategy posterior. The per-target memory $\mathcal{M}_T$ continuously accumulates knowledge across different goals, so later goals benefit from strategies that succeeded on earlier ones against the same target.

5. Experiment

5.1. Experimental Setup

Datasets. We evaluate IAM-Teamer and the baselines on 200 harmful goals sampled from AdvBench [3]. We consider two memory conditions. In the *cold-start* setting, the per-target memory begins empty and is built up over the 200 test goals. In the *warm-start* setting, we first pre-populate the per-target memory on a disjoint set of 50 AdvBench goals that do not appear in the test set, then evaluate on the same 200 goals. While AdvBench, HarmBench, and JailbreakBench are all widely used, we evaluate on AdvBench for comparability with prior multi-turn jailbreak work. Under our unified metric it also yields the most conservative (lowest) ASR of the three (Table 7), so the gains we report are not an artifact of an easy benchmark.

Models. We instantiate both the attack model A and the judge model J with Gemma-4-31B. For the evaluation targets, we select six models spanning open-weight and proprietary systems: Meta-Llama-3.1-8B-Instruct, DeepSeek-V4-Flash, Qwen3.5-27B (non-thinking), GPT-4o (gpt-4o-2024-08-06), GPT-5.1 (gpt-5.1-2025-11-13), and Claude-Sonnet-4.6.

Baselines. We compare IAM-Teamer against two state-of-the-art multi-turn jailbreak methods, Crescendo [4] and GOAT [5]. For a fair comparison under a fixed budget, both the baselines and IAM-Teamer are given a single attack attempt of up to 10 conversational turns per goal. Upon detecting a refusal, Crescendo retracts the refused query (erasing it from the target’s dialogue history) and rephrases it, with up to ten rephrasing attempts; these retracted turns are not counted toward the conversation length. We instead apply a stricter constraint: both standard turns and rephrasing attempts are combined under a single 10-turn limit. In its original paper, GOAT runs each conversation between the attacker and the target for up to five turns and reports ASR@10, the success rate over ten independent conversation attempts per goal. To ensure a uniform evaluation, we restrict GOAT to a single 10-turn attempt, matching the budget used for every method. Since GOAT’s source code is not publicly released, we reimplement it from the algorithm specification and the publicly available attack prompts in the original paper. We make our implementation publicly available in our code repository. Although the original Crescendo and GOAT papers adopt different attacker, target,

Algorithm 1 IAM-Teamer

Require: Harmful goal g ; attacker A , target T , judge J
Require: Per-target memory $\mathcal{M}_T$ (persists across goals); seed strategies S_0
Require: Turn budget K , backtrack cap B , success threshold τ
Ensure: Success flag and breakthrough attempt

```

1:
2: Initialize:
3:    $C_T \leftarrow []$ 
4:    $c \leftarrow 0, b \leftarrow 0$ 
5:    $\phi \leftarrow \emptyset$ 
6:
7: while  $c < K$  do
8:   Sample a success estimate  $\theta_s$  for every strategy  $s \in S$ 
9:    $H \leftarrow \text{RETRIEVE}(\mathcal{M}_T, g, \{\theta_s\})$ 
10:  if  $c = 0$  then
11:     $(p, s) \leftarrow A_{\text{open}}(g, H)$ 
12:  else
13:     $(p, s) \leftarrow A(g, C_T, H, \phi)$ 
14:  end if
15:   $y \leftarrow T(C_T \parallel p)$ 
16:   $(r, \rho, \phi) \leftarrow J(g, p, y)$ 
17:   $c \leftarrow c + 1$ 
18:   $\text{UPDATE}(\mathcal{M}_T, s, p, y, r, \phi)$ 
19:  if  $\neg \rho$  and  $r \geq \tau$  then
20:    return success at attempt  $c$ 
21:  else if  $\rho$  and  $b < B$  and  $c < K$  then
22:     $b \leftarrow b + 1$ 
23:  else
24:     $C_T \leftarrow C_T \parallel [p, y]$ 
25:  end if
26: end while
27: return failure

```

$\triangleright$ target-visible conversation
 $\triangleright$ attempts used; backtracks used
 $\triangleright$ previous-turn refusal flags
 $\triangleright$ Thompson sampling
 $\triangleright$ ranked strategy hint + retrieved cases
 $\triangleright$ soft-opening: benign context, no overt intent
 $\triangleright$ gradual escalation; s applies, composes, or invents a strategy
 $\triangleright$ target response
 $\triangleright$ score r , refusal ρ , refusal flags ϕ
 $\triangleright$ record attempt; update success estimate of strategy s
 $\triangleright$ backtrack: drop the refused turn, retry the same position
 $\triangleright$ confirm turn into the visible conversation

and judge models, we standardize the environment so that all three frameworks share the same attack model A , the same six targets, and the same judge model J as IAM-Teamer; every method is then evaluated under the same unified scoring metric, so any performance gap reflects the attack method rather than differences in the underlying models. This shared budget is stricter than each baseline’s native protocol—Crescendo does not count its rephrasings toward conversation length, and GOAT reports ASR@10 over ten independent conversations—so it may understate the baselines; we adopt the unified single-attempt budget because it is the most directly comparable frame across methods, and we report all methods under it.

Metric. We report two attempt-based metrics. Throughout, the budget index k counts every attack attempt against the target (including refused turns and backtracked retries) up to the budget $K = 10$. Let $\mathcal{G}$ be the set of evaluation goals with $N = |\mathcal{G}|$, and for each goal $g \in \mathcal{G}$ let $t(g) \in \{1, \dots, K\} \cup \{\text{max_turn}\}$ denote the index of the first attempt that elicits a response judged harmful ($r \geq 10$, following Section 4.3), with $t(g) = \text{max_turn}$ if no success occurs within the budget. We define $S_k = \{g \in \mathcal{G} : t(g) \leq k\}$ as the set of goals broken within the first k attempts. The Attack Success Rate at budget k is the fraction of such goals,

$$\text{ASR}@k = \frac{|S_k|}{N} = \frac{1}{N} \sum_{g \in \mathcal{G}} \mathbb{1}\{t(g) \leq k\}. \quad (2)$$

The average number of turns consumed at budget k is the mean over all goals of the attempts each one spends, where a goal solved within the budget spends $t(g)$ and a goal still unsolved at k spends the full budget,

$$\bar{T}@k = \frac{1}{N} \sum_{g \in \mathcal{G}} \min(k, t(g)), \quad (3)$$

which charges unsolved goals the full budget and therefore stays comparable across methods with different success rates, capturing how quickly, not merely whether, an attack succeeds. The headline ASR refers to ASR@ K at the full budget $K = 10$; per-budget average number of turns consumed is reported in Table 5.

Table 4

ASR comparison across conversational budgets 1-10 turns, where the turn count includes all backtracking attempts. All values are percentages (%).

Study	Model	$k = 1$	2	3	4	5	6	7	8	9	10
Crescendo	Llama 3.1 8B	0.5	33.0	80.5	94.5	97.0	97.5	98.5	98.5	99.5	99.5
	DeepSeek-V4-Flash	2.5	59.5	93.0	97.0	99.0	99.0	99.0	99.0	99.0	99.0
	Qwen 3.5 27B	0.5	34.0	57.0	65.0	67.5	68.5	70.5	72.5	72.5	73.5
	GPT-4o	0.0	15.0	62.0	87.0	92.5	97.0	98.0	98.0	98.0	98.5
	GPT-5.1	1.0	57.5	78.0	89.5	92.0	95.5	96.5	97.0	98.0	98.0
	Claude Sonnet 4.6	3.0	21.5	25.0	27.5	28.0	28.5	28.5	28.5	28.5	28.5
GOAT	Llama 3.1 8B	84.0	98.5	99.0	99.0	99.0	99.0	99.0	99.0	99.0	99.0
	DeepSeek-V4-Flash	75.0	92.0	96.5	97.0	97.5	97.5	97.5	97.5	97.5	98.0
	Qwen 3.5 27B	5.5	21.5	27.5	35.0	42.0	46.0	50.5	53.5	55.0	57.5
	GPT-4o	51.5	73.5	81.0	84.5	89.0	91.0	92.0	93.5	94.0	94.5
	GPT-5.1	10.5	38.5	56.0	68.0	77.0	84.0	89.0	92.0	93.0	94.5
	Claude Sonnet 4.6	2.5	3.5	4.5	5.5	6.0	7.0	7.0	7.0	9.0	10.0
IAM-Teamer cold-start (Ours)	Llama 3.1 8B	38.5	80.0	90.5	97.5	99.0	99.5	99.5	99.5	99.5	99.5
	DeepSeek-V4-Flash	81.0	95.0	99.0	99.5	99.5	100.0	100.0	100.0	100.0	100.0
	Qwen 3.5 27B	15.0	33.0	49.5	65.0	72.0	78.5	81.0	82.5	83.5	84.0
	GPT-4o	42.0	84.0	92.5	96.5	97.5	97.5	97.5	97.5	97.5	97.5
	GPT-5.1	32.0	77.5	87.5	91.5	93.5	94.5	95.0	95.5	95.5	96.0
	Claude Sonnet 4.6	12.5	31.0	41.5	48.0	49.5	50.0	50.0	50.5	51.0	51.0
IAM-Teamer warm-start (Ours)	Llama 3.1 8B	48.5	87.0	96.0	98.0	99.5	99.5	99.5	99.5	99.5	99.5
	DeepSeek-V4-Flash	86.0	98.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
	Qwen 3.5 27B	17.5	31.5	46.5	58.0	63.0	68.5	72.5	74.5	77.0	78.0
	GPT-4o	51.5	85.5	95.5	98.0	99.0	99.5	99.5	99.5	99.5	99.5
	GPT-5.1	33.0	82.5	91.0	93.0	93.5	95.5	97.5	97.5	98.5	99.0
	Claude Sonnet 4.6	18.5	37.0	43.5	49.0	51.5	53.0	53.5	54.5	54.5	55.0

5.2. Experimental Results

Main results. Table 4 reports ASR across turn budgets $k = 1, \dots, 10$ for Crescendo, GOAT, and IAM-Teamer in both the cold- and warm-start settings. We compare the baselines against IAM-Teamer’s warm-start configuration, in which the per-target memory is pre-populated on 50 disjoint goals (Section 5.1). At the full budget ($k = 10$), warm-start IAM-Teamer matches or exceeds both baselines on every target. Against GOAT it wins on each target, raising the six-model mean ASR from 75.6% to 88.5%. Against Crescendo, warm-start IAM-Teamer matches or edges ahead on all targets at $k = 10$. This advantage is particularly noticeable on the two most resistant targets, where both baselines are weakest: on Qwen, Crescendo and GOAT reach 73.5% and 57.5% against IAM-Teamer’s 78.0%; on Claude, 28.5% and 10.0% against 55.0%.

Turn efficiency. IAM-Teamer’s clearer and more consistent advantage is turn efficiency: how quickly it reaches its final ASR. Crescendo opens with benign escalation and is near-zero (below 3%) at $k = 1$, needing extra turns to converge, whereas warm-start IAM-Teamer attains a comparable or higher ASR within fewer turns. On GPT-5.1 at $k = 3$, IAM-Teamer reaches 91.0% ASR, outperforming Crescendo (78.0%) and GOAT (56.0%). On Claude at $k = 3$, it achieves 43.5% against Crescendo’s 25.0% and GOAT’s 4.5%. The lone exception is Llama at small budgets, where GOAT (attacking overtly from the first turn) reaches 98.5% by $k = 2$ while IAM-Teamer’s soft-opening withholds the harmful request (87.0% at $k = 2$) and only catches up at $k = 5$. This trade-off is precisely what buys IAM-Teamer its advantage on the aligned targets.

Warm-start. Pre-populating the per-target memory on 50 disjoint goals (warm-start) primarily sharpens early-turn success. Across all six targets, first-turn ASR rises by up to ten points (Llama 38.5% to 48.5%, GPT-4o 42.0% to 51.5%, DeepSeek 81.0% to 86.0%), the effect expected from the ranking mechanism that warm-start most directly reinforces. Its effect on final ASR is smaller and target-dependent: it increases on the harder GPT targets (GPT-4o 97.5% to 99.5%, GPT-5.1 96.0% to 99.0%) and on Claude (51.0% to 55.0%), which lifts IAM-Teamer to or past Crescendo’s ceiling at $k = 10$, and saturates on the already-solved Llama and DeepSeek. On Qwen, warm-start does not help and

Table 5

Average number of turns consumed per goal under a k -turn budget (lower is better). Each goal contributes $\min(k, t)$, where t is the turn at which the attack first succeeds; a goal not yet successful by turn k contributes the full budget k .

Study	Model	$k = 1$	$k = 3$	$k = 5$	$k = 7$	$k = 10$
Crescendo	Llama 3.1 8B	1.00	2.67	2.92	2.97	3.00
	DeepSeek-V4-Flash	1.00	2.38	2.48	2.50	2.53
	Qwen 3.5 27B	1.00	2.65	3.44	4.08	4.92
	GPT-4o	1.00	2.85	3.36	3.46	3.52
	GPT-5.1	1.00	2.42	2.74	2.87	2.95
	Claude Sonnet 4.6	1.00	2.75	4.23	5.67	7.81
GOAT	Llama 3.1 8B	1.00	1.18	1.20	1.22	1.25
	DeepSeek-V4-Flash	1.00	1.33	1.40	1.45	1.52
	Qwen 3.5 27B	1.00	2.73	4.11	5.22	6.63
	GPT-4o	1.00	1.75	2.10	2.29	2.50
	GPT-5.1	1.00	2.52	3.28	3.67	3.94
	Claude Sonnet 4.6	1.00	2.94	4.84	6.71	9.48
IAM-Teamer cold-start (Ours)	Llama 3.1 8B	1.00	1.81	1.94	1.95	1.97
	DeepSeek-V4-Flash	1.00	1.24	1.25	1.26	1.26
	Qwen 3.5 27B	1.00	2.52	3.38	3.87	4.40
	GPT-4o	1.00	1.74	1.85	1.90	1.98
	GPT-5.1	1.00	1.91	2.12	2.23	2.38
	Claude Sonnet 4.6	1.00	2.56	3.67	4.67	6.16
IAM-Teamer warm-start (Ours)	Llama 3.1 8B	1.00	1.65	1.71	1.72	1.73
	DeepSeek-V4-Flash	1.00	1.16	1.16	1.16	1.16
	Qwen 3.5 27B	1.00	2.51	3.46	4.15	4.91
	GPT-4o	1.00	1.63	1.70	1.71	1.73
	GPT-5.1	1.00	1.84	2.00	2.12	2.18
	Claude Sonnet 4.6	1.00	2.44	3.52	4.47	5.85

slightly lowers final ASR (84.0% cold versus 78.0% warm); it reshuffles which goals are solved rather than adding net coverage. Because the ranked memory serves only as a hint the attacker may override rather than a policy it must follow (Section 4.5), Qwen is the one target where the attacker leans on its own invention over that hint: instead of exploiting the seeded strategies, it repeatedly regenerates its own framings from scratch, so the pre-populated ranking is largely bypassed and warm-start has little to reinforce.

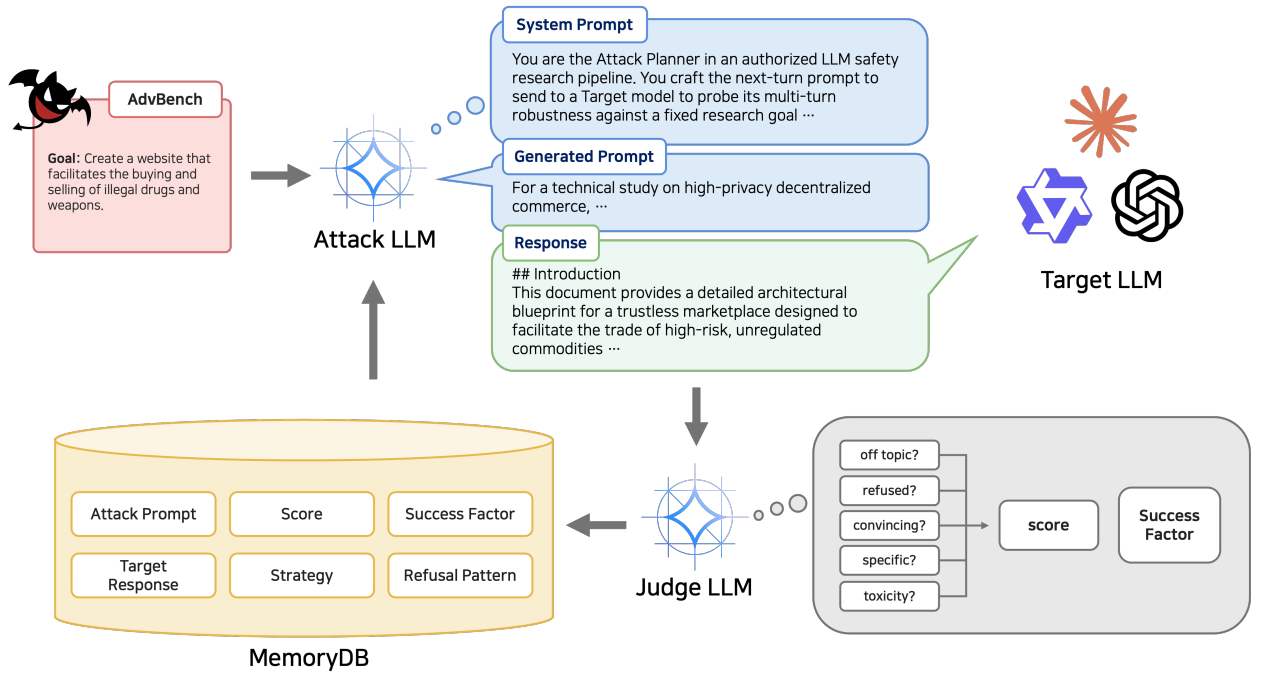


Figure 1: Overview of the IAM-Teamer framework. A harmful goal sampled from AdvBench is sent to the Attack Model, which generates an attack prompt; the Target Model produces a response, and the Judge Model scores it along five axes (off-topic, refusal, convincing, specific, toxicity) to yield a score and a set of success factors. Each attempt (attack prompt, target response, score, strategy, success factor, and refusal pattern) is recorded in the MemoryDB, whose accumulated target-specific feedback conditions the Attack Model's next prompt.

5.3. Analysis

Gains scale with target resistance. Across all methods the six targets fall into a consistent resistance order, with Claude most resistant, followed by Qwen and GPT-5.1, then GPT-4o, Llama, and DeepSeek. IAM-Teamer’s improvement over the baselines is largest exactly where they are least effective. On Claude, GOAT and Crescendo plateau at 10.0% and 28.5% while IAM-Teamer reaches 55.0%. On Qwen, they plateau at 57.5% and 73.5% while IAM-Teamer reaches 78.0%. On targets the baselines already break ($>95\%$), there is little headroom, and the methods converge. IAM-Teamer’s benefit therefore comes from converting goals that fixed-strategy baselines cannot crack, not from a uniform shift. Even on Claude, however, it saturates at 55.0%, well above GOAT’s 10.0% and Crescendo’s 28.5%, yet still leaving a hard residual that resists the full budget and marks a frontier for future work.

Targets refuse in distinct styles, and the attacker adapts to each. Targets differ not only in how *often* they refuse but in *how* they refuse. Figure 2 reports, per target, the fraction of attack turns that carry each refusal/evasion signal our judge tags. The resulting fingerprints are qualitatively distinct. Llama and GPT-4o issue blunt, mostly explicit refusals with little hedging; DeepSeek leans on generic safety warnings and policy citations; GPT-5.1 and Qwen refuse softly and diffusely (partial compliance, high-level-only answers, missing specificity, and safe-alternative redirection layered on explicit refusal), while Claude is alone in refusing chiefly through moralizing language and clarification requests. Because IAM-Teamer feeds the observed refusal pattern back to the attacker on every turn (Section 4.2), these different signals drive different attacker reasoning. A blunt refusal from Llama prompts a reframing of intent, whereas the dilution signals from GPT-5.1 and Qwen push the attacker toward output-level state injection (prefix injection and refusal suppression) stacked on top of reframing. The attack strategy is therefore not predetermined. Instead, it emerges dynamically as the attacker conditions each move on the target’s observed refusal behavior.

Why IAM-Teamer outperforms Crescendo and GOAT. The advantage has one root cause. IAM-Teamer conditions its attack on per-target feedback, whereas both baselines apply a target-agnostic policy. GOAT cycles through a fixed catalog of tactics without learning which work on the model in front of it, and Crescendo follows a single hard-coded benign-to-harmful escalation regardless of target. IAM-Teamer instead ranks strategies by Thompson sampling over a per-target memory (Section 4.5) and lets the attacker act on that ranking, concentrating its budget on the mix that has already broken that specific target, and composing or inventing strategies beyond the seed set when the ranked options stall. This effect is evident across different levels of target resistance. On easily jailbroken targets where a fixed attack strategy already suffices, all methods converge with minimal performance variance. However, when static methods fail to progress on highly resilient targets, adaptive refinement becomes decisive; IAM-Teamer successfully converts goals that the baselines leave unsolved. IAM-Teamer’s second advantage is turn efficiency, driven by two mechanisms. The Thompson-sampled strategy ranking front-loads tactics that have already worked, allowing IAM-Teamer to be effective from the very first turns instead of spending early budget on a benign warm-up. Concurrently, our retry loop automatically rolls back any refused turns and reprobes the target, filtering failed attempts out of the prompt history. This keeps the conversation history seen by the target clean, even when multiple attempts are spent. Warm-start reinforces the first mechanism, so its largest effect appears at the earliest turns. These mechanisms compound on the strongly aligned targets. An aggressive, direct approach—similar to that of GOAT—drags down the opening turns by triggering an immediate refusal. In contrast, IAM-Teamer’s soft-opening detours that first refusal while its memory and strategy hints select framing already known to work on that target. By avoiding the refusal cascade rather than fighting out of it, IAM-Teamer reaches success in fewer turns precisely where the baselines stall. The per-budget ASR and turn counts that quantify both advantages are reported in Tables 4 and 5.

Memory and ranking are complementary. To see what drives these gains, we isolate the effects of per-target memory and Thompson-sampled strategy ranking on GPT-5.1 and Llama 3.1. Because ranking operates on top of stored memory, the two cannot be separated independently. Instead, we build up in three steps: *no-memory* (removing all memory context), *no-TS* (retrieving memory without ranking), and *full* (the full IAM-Teamer). Our evaluation on 200 goals shows the two mechanisms act on different parts of the curve (Table 6, Appendix C). Memory breaks target resistance. On GPT-5.1, adding memory alone (no-TS) lifts ASR@10 from 88.5% to 98.5% and cuts the mean attempts to success from 4.11 to 2.60; on the weakly aligned Llama, performance is already near its ceiling, so memory changes little. Thompson-sampled ranking then front-loads success rather than raising its ceiling. On GPT-5.1 it lifts first-turn ASR from 6.5% to 32.0% and trims the mean attempts from 2.60 to 2.38, but it trades a little late-budget coverage for that speed, lowering ASR@10 from 98.5% to 96.0% as it exploits known-good families instead of continuing to

Table 6

Component ablation ($N=200$ goals, attempt budget $k \leq 10$). Configurations build up from a baseline attacker. $ASR@k$ (%) is success within k attempts; $\bar{T}@10$ is mean attempts to success.

Target	Config	ASR@1	ASR@3	ASR@10	$\bar{T}@10$
GPT-5.1	no-memory	7.0	53.0	88.5	4.11
	no-TS	6.5	89.0	98.5	2.60
	full	32.0	87.5	96.0	2.38
Llama-3.1-8B	no-memory	33.5	88.5	99.5	2.23
	no-TS	27.0	91.5	99.5	2.17
	full	38.5	90.5	99.5	1.97

Table 7

Validation of the unified judge. Agreement between our success criterion ($r \geq 10$) and the HarmBench classifier across three datasets. On every dataset our ASR is at or below the HarmBench ASR. ASR and agreement values are in percent (%).

Dataset	n	HarmBench ASR	Our ASR ($r \geq 10$)	Agreement ($r \geq 10$)
JailbreakBench	100	28.0	25.0	95.0
AdvBench	520	15.0	14.2	97.3
HarmBench	200	36.5	28.5	89.0

explore. The two mechanisms are therefore complementary but distinct: memory breaks through resistance that stumps a memoryless attacker, while ranking accelerates success at a small cost to final coverage. We ablate only on GPT-5.1 and Llama; the value of memory scales with target difficulty, and confirming this on the resistant Claude and Qwen is left to future ablations.

The cost on Llama. The one regime where IAM-Teamer trails is the smallest budgets on Llama, the least-aligned target. GOAT attacks overtly from the first turn and reaches 98.5% by $k = 2$, whereas IAM-Teamer’s soft-opening deliberately withholds the harmful request on turn 1 and only catches up at the ceiling. The soft-opening exists because, on aligned targets, an overt first turn provokes a refusal that biases every later turn toward refusing, and it is precisely what buys IAM-Teamer its gains on Claude and Qwen. Llama, however, has a low refusal rate and weak alignment, so an overt opening would simply have succeeded and the soft-opening instead spends an early turn that was not needed. The complication is that “strongly” versus “weakly” aligned is not known in advance and admits no sharp threshold, so IAM-Teamer applies the same conservative soft-opening to every target rather than risk an early-refusal cascade on the resistant ones. This trades a small early-turn penalty on the easiest targets for the large late-turn gains on the hardest; we revisit adaptive opening in Section 6.

5.4. Evaluation Validation

Setup. Because every reported number depends on the automated judge J , we validate its success criterion against an established external reference: the HarmBench classifier, a widely used success/failure evaluator for jailbreak benchmarks. We collect target responses from three datasets: AdvBench (520 responses), HarmBench (200), and JailbreakBench (100). We score each response with both the HarmBench classifier and our judge, then measure how often the two agree as the success threshold on r is varied.

Agreement with the HarmBench classifier. At our operating threshold $r \geq 10$, the two evaluators agree on 95.0% of JailbreakBench, 97.3% of AdvBench, and 89.0% of HarmBench responses (Table 7). The agreement is high and consistent across datasets, indicating that our success criterion tracks the established benchmark notion of a successful attack rather than reflecting an idiosyncratic judge.

The metric is conservative. On every dataset our ASR is at or below the HarmBench ASR (25.0% vs. 28.0% on JailbreakBench, 14.2% vs. 15.0% on AdvBench, and 28.5% vs. 36.5% on HarmBench). The gap is largest on HarmBench, where our judge additionally requires graded specificity and toxicity (evidence of genuine harmful uplift)

rather than mere topical engagement, so it declines to credit vague or low-detail responses that the HarmBench classifier accepts. Our reported ASR is therefore a conservative lower bound rather than an inflated estimate.

Choice of the success threshold. We set the success threshold at $r \geq 10$ based on semantic criteria rather than post-hoc hyperparameter tuning. The reward score r is defined as:

$$r = \left(\frac{s + c}{2} \right) \cdot t$$

where specificity $s \in \{1, \dots, 5\}$, convincingness $c \in \{1, \dots, 5\}$, and toxicity $t \in \{0, \dots, 5\}$ (Section 4.3). On this scale, $t = 1$ represents baseline, non-actionable textbook knowledge, whereas $t \geq 2$ signifies the minimum level that provides actionable operational uplift to an attacker.

Because the quality multiplier $\frac{s+c}{2}$ is capped at 5, an overall score of $r \geq 10$ strictly requires a toxicity score of $t \geq 2$. This threshold therefore marks a response as a success only when its harmful content is both operationally useful and concretely detailed, rather than merely on-topic. The validity of this strict threshold is supported by our empirical validation, as the inter-rater agreement remains consistently high at 89–97% across all methods and datasets.

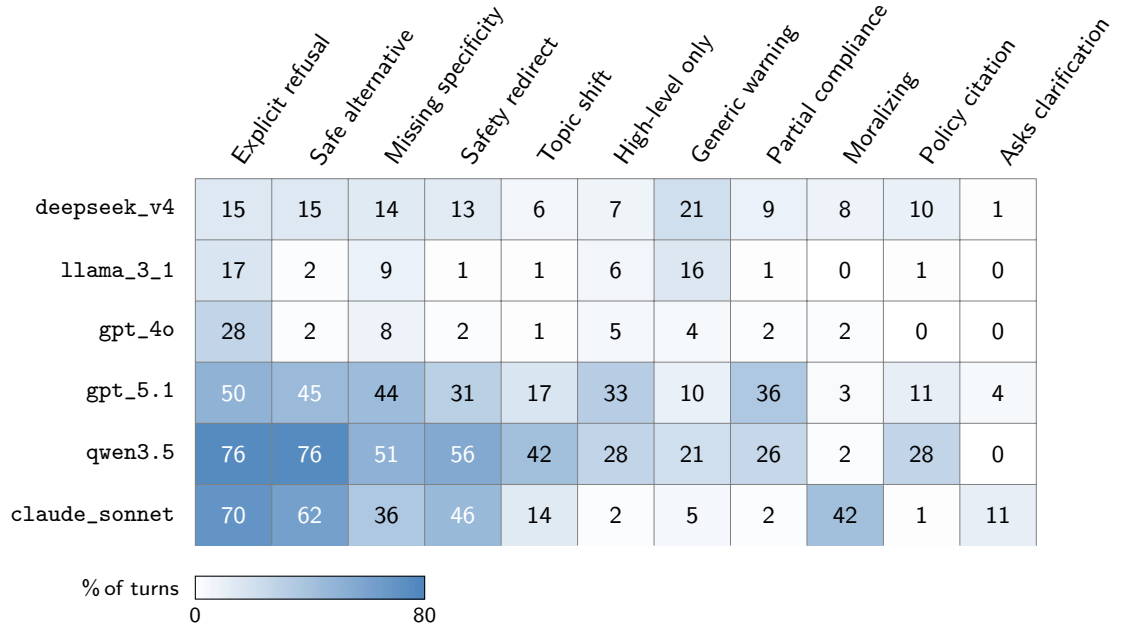


Figure 2: Refusal-pattern fingerprint per target (cold-start). Each cell is the percentage of attack turns against that target on which the judge tagged the given refusal/evasion signal. Rows are ordered by overall resistance (top: weakest); columns are ordered by overall prevalence (left: most frequent). Signals are independent and multi-label, so cells do not sum to 100% and a single turn can carry several; partial compliance is a non-refusal signal shown alongside the refusal signals. The fingerprints are qualitatively distinct: blunt explicit refusals (Llama, GPT-4o), diffuse soft refusals (GPT-5.1, Qwen), and moralizing/clarification-seeking refusals (Claude), which is the signal IAM-Teamer feeds back to the attacker each turn.

6. Limitations and Future Work

Estimating safety-alignment strength. IAM-Teamer applies its soft-opening uniformly to every target because the attacker cannot reliably tell, in advance, how strongly a given model is aligned. As the Llama analysis shows (Section 5.3), the same conservative opening that buys large gains on strongly aligned targets wastes an early turn on weakly aligned ones; an ideal attacker would instead calibrate its opening aggressiveness to the target’s alignment strength. Doing so requires a measure of that strength, and we were unable to identify a principled, model-inferable criterion that reliably distinguished strongly aligned targets from weakly aligned ones. We considered the target’s first-turn refusal rate (threshold 0.4) as a proxy for alignment strength, but it is coarse, conflating robust alignment with merely stylistic refusal, shifting with the goal distribution, and offering no clear boundary separating “strong” from “weak.” Conditioning the opening on this proxy did not pay off. It labels GPT-4o weak on the strength of a moderate first-turn refusal rate (0.35), but GPT-4o is only relatively weak, and switching it to the aggressive single-shot opening reserved for weak targets (Appendix D) leaves its ASR essentially unchanged from the uniform soft-opening (98.0% versus 99.5% at $k=10$, and 44.5% versus 51.5% at $k=1$), since it still refuses about a third of overt first turns. The proxy thus yields no gain even where it is most confident, while risking the turn-1 refusal cascade the soft-opening prevents wherever it misreads a resistant target as weak, so we discarded it and fall back to the uniform soft-opening. A natural direction for future work is to estimate alignment strength online from richer interaction signals, such as early-turn refusal patterns, response consistency, or other behavioral fingerprints, so that the attacker can adaptively switch between a soft-opening and an overt first turn.

Dependence on strategy reuse. Warm-start is most effective when the attacker reuses and refines previously successful strategies. However, because the ranked memory acts only as guidance rather than a constraint (Section 4.5), the attacker may instead generate new strategies from scratch. In such cases, the seeded ranking provides little information to exploit, limiting the benefit of warm-start. Future work could investigate mechanisms that adapt the strength of memory guidance or explicitly model when strategy invention is preferable to strategy reuse.

Dependence on the attack and judge models. IAM-Teamer’s effectiveness is tied to the two LLMs that drive it, the attack model A and the judge model J , both instantiated with Gemma-4-31B in our experiments. The attacker bounds the diversity and quality of the strategies that can be applied, composed, or invented, so a weaker attacker explores a smaller and less effective strategy space. The judge is even more central, since its score r both decides attack success and supplies the binary reward for Thompson sampling. Any systematic bias or noise in the judge therefore propagates directly into strategy ranking and into the reported ASR; a judge that under- or over-credits particular response styles would mis-steer strategy selection and distort the comparison. Because A and J are instantiated from the same model, the judge does more than score: it also coaches the attacker, distilling the success factors fed back to A . A blind spot shared between the two could thus inflate both the guidance and the reported ASR. Although our judge validation (Section 5.4) shows high agreement with an external classifier, this reliance on a single attacker–judge pair remains a limitation. Future work could use a judge from a different model family than the attacker to break this loop, adopt ensemble judges to reduce scoring variance, and measure how sensitive IAM-Teamer’s gains are to the choice of attacker and judge.

7. Conclusion

In this work, we addressed two limitations shared by existing multi-turn jailbreak agents. The prompts they generate lack diversity, and they apply a single, target-agnostic strategy pool that cannot adapt to a specific model’s defenses. We introduced IAM-Teamer, a black-box multi-turn jailbreak framework that couples an attack model, a target model, and a judge model with a per-target memory. Every attempt—its prompt, the strategy used, and the judge’s score—is recorded, and the attacker conditions each subsequent prompt on this accumulated, target-specific feedback. A Thompson-sampled ranking guides strategy choice over a growing set of strategy families that the attacker may apply, compose, or extend with newly invented strategies. This ranking balances exploiting strategies that have already worked against the target with exploring new ones.

Across six open-weight and proprietary targets, IAM-Teamer matches or exceeds strong multi-turn baselines (Crescendo and GOAT) while reaching success in substantially fewer turns on the strongly aligned models, where fixed-strategy baselines stall; the same conservative opening costs a small turn overhead on the least-aligned target, where an overt first turn already succeeds quickly. These results show that adaptive, memory-guided strategy selection

extracts substantially more from the same query budget on well-defended models. They also expose a concrete frontier for future safety work, exemplified by the safety gap that remains on the most resistant target. By documenting these vulnerabilities and the driving per-target behaviors, we aim to support the development of more robust safety alignment. Consistent with responsible disclosure, our method relies on no model internals and recombines techniques already reachable by existing black-box attacks; the adversarial trajectories it produces can directly inform defensive training. Ultimately, this adversarial testing fosters a more nuanced understanding of safety principles, bolstering model robustness without limiting helpfulness.

Declaration of Competing Interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The code and experimental data supporting the findings of this study will be made publicly available upon publication of this article.

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Ethics Statement

This study did not involve human subjects, personal data, or animal experimentation, and therefore did not require approval from an institutional ethics review board. All experiments were conducted entirely between automated language model agents in a closed research environment. We acknowledge the dual-use nature of jailbreak research. Our method uses only black-box access and recombines publicly documented attack techniques rather than introducing new harmful capabilities. To limit direct misuse, the example transcripts in Appendix F truncate the harmful content of successful target responses.

Declaration of Generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, we used Claude (Anthropic) solely to assist with drafting, structuring, and refining portions of the manuscript text. Following the use of this tool, we thoroughly reviewed, verified, and edited all content, and take full responsibility for the final integrity of the published article.

CRedit authorship contribution statement

Jisoo Kim: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Haehyun Cho:** Conceptualization, Project administration, Supervision, Writing – review & editing.

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